

Grade 6 Accelerated
Unit 5
Lesson 4 Homework

Name Answer Key
Date _____
Period _____

Carrie is starting a dog-walking business. The table represents her first three jobs.

Miles of Walking	One Dog	Two Dogs	Three Dogs
1.5 miles	\$6		
2 miles			\$16
2.5 miles		\$15	

1. What does Carrie charge per one mile of dog-walking for one dog?

$$\frac{\$6}{1.5 \text{ mi}} = \$4 \text{ per mile for one dog}$$

2. What is the unit rate for walking two dogs?

$$\frac{\$15}{2.5 \text{ mi}} = \$6 \text{ for two dogs per mile}$$

3. When Carrie walks three dogs, what does she charge per mile?

$$\frac{\$16}{2 \text{ mi}} = \$8 \text{ per mile}$$

4. How much does the unit rate increase for each additional dog?

\$2

5. Carrie wants to purchase a pair of running shoes for \$75. She will need to walk 12.5 miles for the family with two dogs to purchase the shoes.

$$\frac{\$75}{\$6} = 12.5 \text{ miles}$$

6. A family has hired Carrie to walk their three dogs every Tuesday and Thursday afternoon. If they agree to pay her \$36 every week, Carrie will walk the dogs

- ☐ 4
☐ 4.5

miles each week.

$$\frac{\$8}{\text{mile}} = \$36 \text{ for 4.5 miles}$$

Carrie has determined she needs a pricing sheet to be more efficient when quoting prices to dog owners.

Miles of Walking	One Dog	Two Dogs	Three Dogs
1 mile	\$4	\$6	\$8
1.5 miles	\$6	\$9	\$12
2 miles	\$8	\$12	\$16
2.5 miles	\$10	\$15	\$20
3 miles	\$12	\$18	\$24

7. Fill in the table by using the unit rates you determined previously.

Carrie has taken a partner to expand the dog-walking business. They have combined all money earned for one week to pay for business cards.

8. Carrie walked three dogs 4.5 miles, two dogs 3 miles, and one dog 5 miles during the week. Carrie earned ☐ \$69 ☒ \$74 to put towards business cards.

$$(8 \cdot 4.5) + (6 \cdot 3) + (4 \cdot 5) = 36 + 18 + 20 = 74$$

9. Her partner walked one dog 2.5 miles, two dogs 4 miles, and walked three dogs 2 miles. Carrie's partner earned ☐ \$50 ☐ \$46 to put towards business cards.

$$(4 \cdot 2.5) + (6 \cdot 4) + (8 \cdot 2) = 10 + 24 + 16 = 50$$

10. Carrie will have enough money to purchase

- ☒ 500 glossy business cards for \$110.
- ☐ 1000 glossy business cards for \$150.

$$74 + 50 = \$124$$